

Identify patterns in synthesis, decomposition and displacement reactions and investigate the factors that affect reaction rates

Learning intention

- identify patterns in synthesis, decomposition and displacement reactions and investigate the factors that affect reaction rates.

Teach and model

- identifying reaction type and predicting the products.
- examining reactions that are used to produce a range of useful products.
- investigating synthesis reactions such as reaction of metals with oxygen.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.